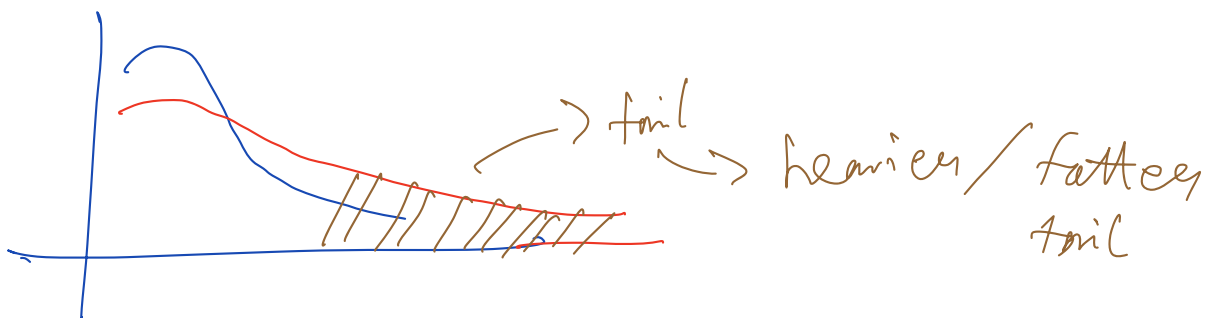


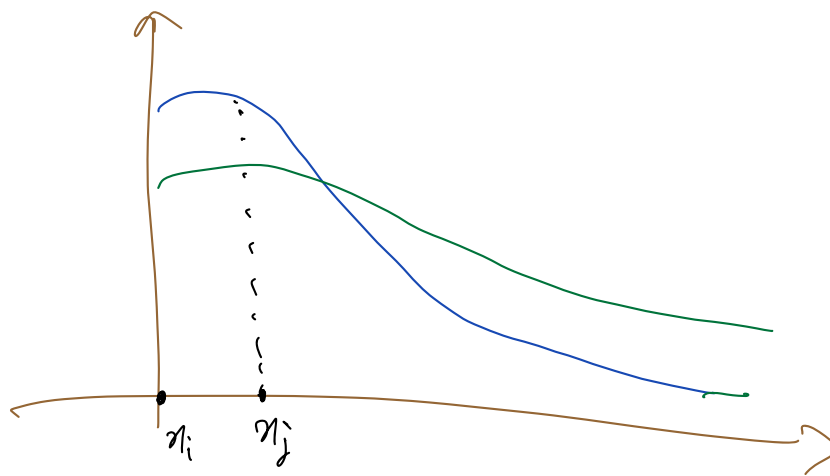
Cont. lecture (Oct 25)

- 1) PCA Recap
- 2) Information loss in PCA
- 3) Code
- 4) High-dim viz using PCA
- 5) t-SNE
- 6) Problems with t-SNE

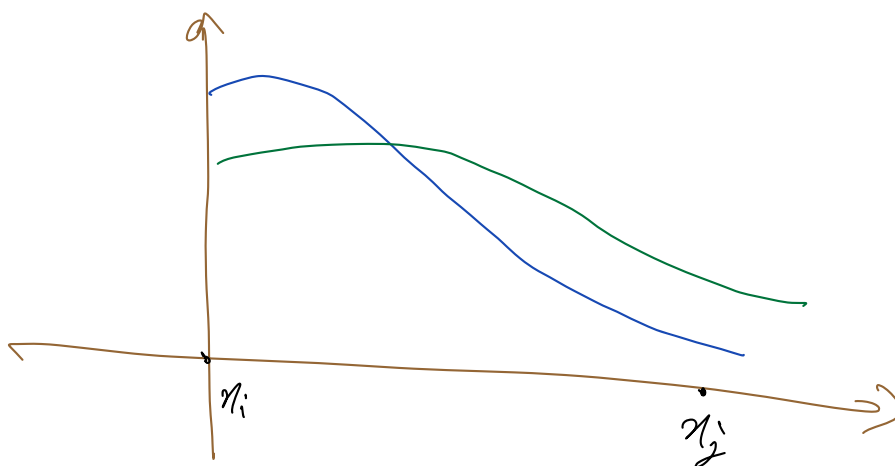
Today's lecture

- 1) Recap of PCA
- 2) t-SNE quick recap
- 3) prob. for higher dim
- 4) prob. for lower dim
- 5) implementation of t-SNE
- 6) UMAP (if time permits)





Tip



Bottom

$$\begin{bmatrix} x_1, y_1, z_1 \\ x_2, y_2, z_2 \\ \vdots \\ x_{300}, y_{300}, z_{300} \end{bmatrix}$$

$$\begin{bmatrix} x'_1, y'_1 \\ x'_2, y'_2 \\ \vdots \\ x'_{300}, y'_{300} \end{bmatrix}$$

$\varepsilon 38$

$$(300 \times 2)$$

600 learnable params